

CURRICULUM VITAE (with Publication List)

(Cüneyt F. Bazlamaçcı)

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Major Interests and Research Areas:

Communication and Computer Networks

- *Synthesis and design*
- *Architectures and protocols*
- *Fast packet switching and processing systems*

Computer Systems Architecture and Hardware

- *Microprocessor systems and applications*
- *Reconfigurable systems*
- *Embedded systems*

Optimization, Graph Algorithms and Applications

- *Optimization theory and mathematical programming*
- *Algorithms and advanced data structures*

Education:

Degree	Department/Program	University	Year
PhD	Department of Computation	University of Manchester Institute of Science and Technology (UMIST)*	Dec. 1996
MSc	Electrical and Electronics Engineering Department	Middle East Technical University	July 1991
BSc	Electrical and Electronics Engineering Department	Middle East Technical University	July 1988
High School Diploma	Mathematics Branch	Ankara Atatürk Anadolu High School	June 1984

* UMIST and Victoria University of Manchester are merged in 2004 and named as University of Manchester and Department of Computation is merged with School of Computer Science.

PhD Thesis and Supervisor:

Bazlamaçcı, C.F., "Optimized network design: minimum spanning trees and minimum concave-cost problems, PhD Thesis, University of Manchester Institute of Science and Technology (UMIST), December 1996. Supervisor: Prof. Dr. Khalil S. Hindi

MSc Thesis and Supervisor:

Bazlamaçcı, C.F., "A fast Fourier transform implementation and performance evaluation on a transputer based parallel computer system", MSc Thesis, Middle East Technical University, July 1991. Supervisor: Prof. Dr. Hasan Güran

Employment History:

Job Title	Place	Year
Full-time faculty (Prof. Dr.)	Izmir Institute of Technology, Computer Eng. Dep., İzmir, Turkey	2020-
Full-time faculty (Assoc. Prof. Dr.)	Middle East Technical University, Electrical and Electronics Eng. Dep., Ankara, Turkey	2010-2020
Visiting faculty (Asst. Prof. Dr.)	Middle East Technical University Northern Cyprus Campus, Electrical and Electronics Eng. Dep., Kalkanlı, Guzelyurt, TRNC (Visiting Scholar)	Feb '09 – June '09
Full-time faculty (Asst. Prof. Dr.)	Middle East Technical University, Electrical and Electronics Eng. Dep., Ankara, Turkey	1997-2010
Military Service	Media Department, General Plan and Principles Div., Land Forces, Ankara, Turkey	1998
Part-time Researcher	Communication Systems and Data Networks Research Group, TUBITAK-BILTEN Information Technologies and Electronics Institute, Ankara, Turkey	1997-1998
Full-time faculty (Instr. Dr.)	Middle East Technical University, Electrical and Electronics Eng. Dep., Ankara, Turkey	1996-1997
Teaching Assistant	University of Manchester Institute of Science and Technology (UMIST), Computation Department, Manchester, England	1992-1996
Research Assistant	Middle East Technical University, Electrical and Electronics Eng. Dep., Ankara, Turkey (on leave to pursue PhD at UMIST)	1992-1996
Research Assistant	Middle East Technical University, Electrical and Electronics Eng. Dep., Ankara, Turkey	1990-1992
R&D Engineer	Aselsan Military Electronics Inc., Control Systems Dep., Ankara, Turkey	1988-1990
Technical Staff	Control Systems Dep., Communications Systems Dep., Ankara, Turkey	1986-1988

Date of Associate Professorship: 28 May 2010

Science Area: Electrical and Electronics Engineering

Key subtopics: i) Communication and Computer Networks
ii) Computer Systems Architecture and Hardware

For a summary of research results (in categorized format), please visit

<https://web.iyte.edu.tr/%7Ecuneytbazlamacci/research%20results%20categorized.html>

PUBLICATIONS**Patents (International):**

Bazlamaççı, C.F. and Erdem, O. "Systolic array architecture for fast IP lookup" (triadic patent achieved under METU Technopark portfolio)

- European Patent No: EP2517420B1, Date of patent: March 19, 2014
- USA Patent No: US 8,724624 B, Date of patent: May 13, 2014
- Japanese Patent No: 5529976, Date of patent: April 25, 2014

Patent (National):

Bazlamaççı, C.F. and Erdem, O. "Hızlı IP arama için sistolik dizilim mimarisi"

- TPE Patent No: TR 2012 09797 B, Date of patent: March 19, 2014

Journal Articles (International) (SCI and SCIE):

Kocaay, B. and **Bazlamaççı, C.F.** "Semi-Autonomous Polyhedral Optimization Framework for HLS-Based SoC Acceleration", Paper in review, submitted to *PeerJ Computer Science* (2026). [Q1]

Bekdemir, L. and **Bazlamaççı, C.F.** "Hybrid probabilistic timing analysis with extreme value theory and copulas", *Microprocessors and Microsystems*, 89, Article No. 104419, (2022). [Q3]

Eylen, T. and **Bazlamaççı, C.F.** "One-way active delay measurement with error bounds", *IEEE Trans. on Instrumentation and Measurement*, 64/12, 3476-3489 (2015). [Q1]

Çelik, C. and **Bazlamaççı, C.F.** "Evaluation of energy and buffer aware application mapping for networks-on-chip", *Microprocessors and Microsystems*, 38/4, 325-336, (2014). [Q3]

Çelik, C. and **Bazlamaççı, C.F.** "Energy and buffer aware application mapping for network-on-chips with self-similar traffic", *Journal of Systems Architecture*, 59/10, 1364-1374, (2013). [Q1]

Erdem, O. and **Bazlamaççı, C.F.** "High performance IP lookup engine with compact clustered trie search", *Computer Journal*, 55/12, 1447-1466, (2012). [Q2]

Say, F. and **Bazlamaççı, C.F.** "A reconfigurable computing platform for real time embedded applications" *Microprocessors and Microsystems*, 36/1, 13-32, (2012). [Q3]

Erdem, O. and **Bazlamaççı, C.F.** "Array design for trie-based IP lookup" *IEEE Communications Letters*, 14/8, 773-775, (2010). [Q2]

Cinel, S. and **Bazlamaççı, C.F.** "A distributed heuristic algorithm for the rectilinear Steiner minimal tree problem", *IEEE Trans. on Computer-Aided Design of Integrated Circuits and Systems*, 27/11, 2083-2087 (2008).

Bazlamaççı, C.F. and Say, F. "Minimum concave cost multicommodity network design", *Telecommunication Systems*, 36/4, 181-203 (2007).

Bazlamaççı, C.F. and Hindi, K.S. "Minimum-weight spanning tree algorithms: A survey and empirical study", *Computers & Operations Research*, 28/8, 767-785 (2001).

Bazlamaççı, C.F. and Hindi, K.S. "Enhanced adjacent extreme point search and Tabu search for the minimum, concave-cost uncapacitated transshipment problem", *Journal of the Operational Research Society*, 47/9, 1150-1165 (1996).

Conference Articles in Proceedings (International):

Alasmar, M. and **Bazlamaççı, C.F.**, "Workload distribution on heterogeneous platforms", *Proc. of the IEEE Int. Conf. on Computer, Information, and Telecommunication Systems (CITS 2021)*, İstanbul (Turkey), 149-153, (2021).

Sert, K. and **Bazlamaççı, C.F.**, "NFA Based Regular Expression Matching on FPGA", *Proc. of the IEEE Int. Conf. on Computer, Information, and Telecommunication Systems (CITS 2021)*, İstanbul (Turkey), 144-148, (2021).

Soltani, A. and **Bazlamaççı, C.F.**, "HyFI: Hybrid flow initiation in software defined networks", *Proc. of the 5th Int. Conf. on Information and Communication Systems (iCICS 2014)*, Irbid (Jordan) 1-3 April 2014.

Alizadeh, E.J. and **Bazlamaççı, C.F.** "Life time sensitive weighted clustering on wireless sensor networks", *Proc. of the 3rd Int. Conf. on Sensor Networks (SENSORNETS 2014)*, Lisbon (Portugal), 41-51 (2014).

Yıldırım, S. and **Bazlamaçcı, C.F.** "A true random number generator and test platform built in FPGA", *Proc. of the 5th Int. Conf. on Information Security and Cryptology (ISCTurkey 2012)*, Ankara (Turkey), 262-267 (2012).

Çelik, C. and **Bazlamaçcı, C.F.** "Effect of application mapping on network-on-chip performance", *Proc. of the 20th Euromicro Int. Conf. on Parallel, Distributed and Network-Based Computing (PDP 2012)*, Munich (Germany), 465-472 (2012).

Uğurel, G. and **Bazlamaçcı, C.F.** "Context switching time and memory footprint comparison of Xilkernel and μ C/OS-II on MicroBlaze", *Proc. of the 7th Int. Conf. on Electrical and Electronics Engineering (ELECO 2011)*, Bursa (Turkey), v.2, 52-55 (2011).

Erdem, O., Hoang, L., Prasanna, V.K. and **Bazlamaçcı, C.F.** "Hybrid data structure for IP lookup in virtual routers using FPGAs", *Proc. of the 22nd IEEE Int. Conf. on Application-specific Systems, Architectures and Processors (ASAP 2011)*, ed. J.R. Cavallaro, et. al., Santa Monica, California (USA) 95-101, IEEE (2011).

Erdem, O. and **Bazlamaçcı, C.F.** "MIPS extension for a TCAM based parallel architecture for fast IP lookup", *Proc. of the 24th Int. Symp. on Computer and Information Sciences (ISCIS 2009)*, ed. F.Y. Vural, et. al., Güzelyurt (Northern Cyprus Turkish Republic), 310-315, IEEE (2009).

Bayramoğlu, N.Y. and **Bazlamaçcı, C.F.** "Recursive shortest spanning tree algorithms for image segmentation", *Proc. of the IEEE International Conference on "Computer as a Tool" (EUROCON 2005)*, Belgrade (Serbia and Montenegro), 979-982, IEEE Press (2005).

Çelik, C. and **Bazlamaçcı, C.F.** "Performance analysis of reliable multicast protocols", *Proc. of the 20th Int. Symp. on Computer and Information Sciences (ISCIS 2005)*, İstanbul (Turkey), Advances in Computer Science and Engineering: Reports – Vol. 1: New Trends in Computer Networks, ed. T.Tugcu, et. al., 319-328, Imperial College Press, London, UK (2005).

Öztürk, E.A. and **Bazlamaçcı, C.F.** "Fairness in differentiated services architecture", *Proc. 17th Int. Symp. on Computer and Information Sciences (ISCIS XVII)*, ed. I. Çiçekli, et. al., Florida (USA), 264-268, CRC Press (2003).

Önem (Ertem), D. and **Bazlamaçcı, C.F.** "Minimum weight directed spanning tree problem with degree, hop and capacity constraints", *Proc. 16th Int. Symp. on Computer and Information Sciences (ISCIS XVI)*, ed. S. Kuru, et. al., Antalya (Turkey), 429-436, Işık University (2001).

Bazlamaçcı, C.F. and Hindi, K.S. "An empirical study of minimum-weight spanning tree algorithms, classical and modern," *Proc. of the 14th Int. Symp. on Computer and Information Sciences (ISCIS XIV)*, ed. M. Türksever, et. al., Kuşadası (Turkey), 370-380, Ege University (1999).

Bazlamaçcı, C.F. and Hindi, K.S. "Verifying minimum spanning trees in linear time," *Proc. of the Symp. on Operations Research (SOR 1996)*, ed. U. Zimmerman et. al., Braunschweig (Germany), 139-144, Springer-Verlag, Heidelberg (1997).

Bazlamaçcı, C.F. and Hindi, K.S. "A tabu search algorithm for concave cost transshipment problems," *Proc. of the 3rd IFIP WG-7.6 Working Conference on Optimization-based Computer-aided Modeling and Design (IFIP 1994)*, ed. J. Dolezal and J. Fidler, Prague (Czech Rep.) 94-101, Academy of Sciences of the Czech Rep. (1994).

Güran, H., and **Bazlamaçcı, C.F.** "Performance of parallel FFT on a transputer based mesh architecture," *Proc. of the 7th Int. Symp. on Computer and Information Sciences (ISCIS VII)*, ed. E. Gelenbe, et. al., Antalya (Turkey), 503-506, Universite Rene Descartes (1992).

Conference Articles in Proceedings (National):

Akyüz, G. and **C.F. Bazlamaçcı**, "Sleep-Enabled LoRaWAN Gateway via Spreading-Factor Clustering (Yayılım Faktörü Kümeleme ile Uyku Destekli LoRaWAN Ağ Geçidi)", *Proc. of the 34th*

Signal Processing and Communications Applications Conf. (SIU 2026), İstanbul (Turkey), Piri Reis University, 7-10 July 2026. (in Turkish)

Bekdemir, L. and **C.F. Bazlamaççı**, "Hybrid Probabilistic Timing Analysis (Hibrit olasılıksal zamanlama analizi)", *Proc. of the 15th Nat. Symp. on Software Engineering (UYMS'21)*, İzmir (Turkey), November 17-19, 2021. (in Turkish)

Tengilimoğlu, B. and **C.F. Bazlamaççı**, "Partial reconfiguration on a real-time target detection and tracking system", *Proc. of the 22nd Signal Processing and Communications Applications Conf. (SIU 2014), Trabzon (Turkey)*, Karadeniz Technical University, 23-25 April 2014. (in Turkish)

Erdem, O. and **C.F. Bazlamaççı**, "SRAM based systolic array architecture for fast IP lookup and matching", *Proc. of the 13th. Nat. Symp. on Electrical, Electronics, Computer and Biomedical Engineering, Ankara (Turkey)*, 173-178, EMO (2009). (in Turkish)

Kahraman, E., T.İpek, B.İyidir, **C.F. Bazlamaççı** and S.Bilgen, "A domain engineering study towards developing a component based product line", *Proc. of the 4th Nat. Symp. on Software Engineering (UYMS'09)*, İstanbul (Turkey), 283-288, EMO (2009). (in Turkish)

Bazlamaççı, C.F. and B. Karadeniz, "Lower bounds for the minimum span frequency assignment problem in cellular networks", *Proc. of the Electrical Electronics and Computer Engineering Symposium (ELECO'02)*, Bursa (Turkey), Electronics Volume: 252-256 TMMOB, Uludağ University, Istanbul Technical University, Tübitak (2002). (in Turkish)

Bazlamaççı, C.F., "Strategic information warfare and national information security", *Proc. of the 8th Electrical Electronics and Computer Engineering Symposium, Gaziantep (Turkey)*, Vol. I, 1-4, Gaziantep University, EMO, Tübitak (1999). (in Turkish)

Bazlamaççı, C.F. and K.S. Hindi, "A branch and bound algorithm for the capacitated minimal spanning tree problem using a directed-flow formulation," *Proc. of the 2nd Symp. on Computer Networks (BAS'97)*, ed. S. Bilgen and A. Özgüt, Ankara (Turkey), 31-40, Middle East Technical University (1997).

Ünal, E., **Bazlamaççı, C.F.**, and Güran, H., "Investigation of transputer performance using transform coding of images and FFT methods", *Proc. of the Electrical Electronics and Computer Engineering Conference (BİLKON'91)*, Ankara (Turkey), 270-274, Bilkent University (1991). (in Turkish).

Other Conference Articles in Proceedings (International) (Abstract only):

Erdem, O., Hoang, L., Prasanna, V.K. and **Bazlamaççı, C.F.** "Hybrid data structure for IP lookup in virtual routers using FPGAs", *19th IEEE Int. Symp. On Field-Programmable Custom Computing Machines (FCCM 2011)* (poster session), Salt Lake City, Utah (USA), Abstract Volume p.253 (2011).

Bazlamaççı, C.F. and Hindi, K.S. "Disaggregate local search for minimum concave-cost multi-commodity network design," *Proc. of the Symp. on Operations Research (SOR 1997)*, Jena (Germany), Abstract Volume p.46 (1997).

Bazlamaççı, C.F. and Hindi, K.S. "A branch and bound algorithm for the capacitated minimal spanning tree problem using a directed formulation," *Proc. of the 16th Int. Symp. on Mathematical Programming (ISMP 1997)*, Lausanne (Switzerland), Abstract Volume p.43 (1997).

Other misc. publications:

Bazlamaççı, C.F. (editor) and Demir, Ş. *Electrical Electronics Computer Engineering Education 1st Nat. Symp. Proceedings*, METU, EMO, Tubitak, Ankara (2003).

Uğurel, G. and **C.F. Bazlamaççı**, "Comparison of real time operating systems on MicroBlaze" *Automation (Otomasyon)*, Vol. 236, 316-322, Bileşim Yayıncılık, (2012). (in Turkish)

Supervised PhD Theses:

Çelik, C., "*Energy and buffer aware application mapping for network on chips*", PhD Thesis, Middle East Technical University, February 2013.

Erdem, O., "*Parallel and pipelining architectures for high speed packet forwarding*", PhD Thesis, Middle East Technical University, September 2011 (*recipient of the "Thesis of the Year Award 2010-2011" by METU Graduate School of Natural and Applied Sciences*).

Say, F., "*A reconfigurable computing platform for real time embedded applications*", PhD Thesis, Middle East Technical University, September 2011.

Supervised MSc Theses:

(ongoing) Akyüz, G., "*Integration of vehicle-to-everything (V2X) communication in infotainment systems*", MSc Thesis, IZTECH, (expected graduation time) July 2027.

(ongoing) Huseynzade, M., "*Fragmentation aware GPU scheduling for lightweight virtualization based distributed systems*", MSc Thesis, IZTECH, (expected graduation time) July 2027.

(ongoing) Aslanca, K., "*Fairness and stability in heterogeneous congestion control environments*", MSc Thesis, IZTECH, (expected graduation time) July 2027.

(ongoing) Erdal, B., "*Epigenetic-Inspired Mechanisms for Improving Evolutionary Optimization Algorithms*", MSc Thesis, IZTECH, (expected graduation time) July 2027.

Dönüm, N., "*Ownership-inspired typestate analysis for memory safety in C/C++ code with IPC primitives*", MSc Thesis, IZTECH, July 2026.

Alasmar, M., "*FPM based partitioning and assignment algorithm for data parallel applications on heterogeneous platforms*", MSc Thesis, METU, July 2022.

Kocaay, A.B., "*Boosting performance of HLS optimization for SoC based hardware accelerators*", MSc Thesis, METU, January 2020.

Toka, N.A., "*Design and implementation of power quality monitoring system for distribution systems*", MSc Thesis, METU, September 2019.

Yılmaz, E., "*Feasibility study for dynamic context switching in partially reconfigurable FPGAs*", MSc Thesis, METU, September 2019.

Cilasın, R., "*An application-aware DRAM controller*", MSc Thesis, METU, September 2019.

Bekdemir, L., "*Hybrid probabilistic timing analysis with extreme value theory and copulas*", MSc Thesis, METU, September 2019.

Uslu, Ç., "*Optimizing core signal processing functions on a superscalar SIMD architecture*", MSc Thesis, METU, August 2019.

Sert, K., "*NFA based regular expression matching on FPGA*", MSc Thesis, METU, September 2018.

Gözütok, A., "*Data plane-based defense system against DDOS attacks for software defined networks*", MSc Thesis, METU, March 2018.

Efe, O., "*Tool support for worst case end to end delay analysis of AFDX networks*", MSc Thesis, METU, February 2016.

- Özcan, D., "*Parallel rectilinear Steiner tree solution using GPGPU*", MSc Thesis, METU, December 2015.
- Eylen, T., "*One way active delay measurement with error bounds*", MSc Thesis, METU, August 2015.
- Kocalar, G., "*GF(2^m) multiplier implementation on a partially reconfigurable FPGA*", MSc Thesis, METU, May 2015.
- Dikbayır, H.S., "*A hybrid resource allocation system for cloud computing*", MSc Thesis, METU, September 2014.
- Büyüksahin, M.U., "*OpenCL implementation of Montgomery multiplication on FPGA*", MSc Thesis, METU, September 2014.
- Tengilimoğlu, B., "*Partially reconfigurable FPGA implementation of a real-time system*", MSc Thesis, METU, June 2014.
- Özer, C., "*A dynamic memory manager for FPGA applications*", MSc Thesis, METU, June 2014.
- Soltani, A., "*Flow initiation in software defined networking*", MSc Thesis, METU, February 2014.
- Baysal, D., "*Trie-tree data structure for IP lookup in virtual routers*", MSc Thesis, METU, January 2014.
- Jarchlo, E.A., "*Life-time sensitive weighted clustering algorithm on wireless sensor networks*", MSc Thesis, METU, September 2013.
- Rumelili, C.I., "*Memory organization in pipelined hierarchical search structures for packet classification*", MSc Thesis, METU, June 2013.
- Yıldırım, S., "*A true random number generator in FPGA for cryptographic applications*", MSc Thesis, METU, December 2012.
- Uğurel, G., "*Benchmarking of Xilkernel, FreeRTOS and µC/OS-II on the soft processor platform MicroBlaze*", MSc Thesis, METU, June 2012.
- Özkaner, A., "*Prototype development and verification of an IP lookup engine on FPGA*", MSc Thesis, METU, February 2012.
- Kılınç, I.O., "*FPGA implementation of a network-on-chip*", MSc Thesis, METU, September 2011.
- Akbaş, M.I., "*Evaluation of core stateless guaranteed fair network architecture*", MSc Thesis, METU, September 2006.
- Cinel, S., "*Sequential and parallel heuristic algorithms for the rectilinear Steiner tree problem*", MSc Thesis, METU, September 2006.
- Say, F., "*Minimum concave cost multicommodity network design problem*", MSc Thesis, METU, September 2005.
- Bereketli, A., "*On QoS multicast routing protocols*", MSc Thesis, METU, September 2005.
- Bayramoğlu, N.Y., "*Recursive shortest spanning tree algorithms for image segmentation*", MSc Thesis, METU, July 2005.
- Çelik, C., "*Performance analysis of reliable multicast*", MSc Thesis, METU, December 2004.

Özbilgin, G., "Reroute sequence planning in multiprotocol label switching networks", MSc Thesis, METU, December 2004.

Öğünlü, B., "Lifetime analysis for wireless sensor networks", MSc Thesis, METU, August 2004.

Önal, K., "Internet multicast congestion control", MSc Thesis, METU, February 2004.

Kaçar, M.N., "A soho router implementation on Motorola MCF5272 processor and UCLinux operating system", MSc Thesis, METU, August 2003.

Yeler, T., "University course scheduling using a genetic algorithm", MSc Thesis, METU, December 2002.

Yılmaz, H., "Traveling salesman problem: Solution with branch and bound and data correction algorithms", MSc Thesis, METU, December 2001.

Karadeniz, B., "Lower bounds for the minimum span frequency assignment problem in cellular networks", MSc Thesis, METU, December 2001.

Öztürk, E., "Fairness in differentiated services architecture", MSc Thesis, METU, December 2001.

Ertem (Önem), D., "Minimum weight directed spanning tree problem with degree, hop and capacity constraint", MSc Thesis, METU, April 2001.

Alumni:

PhD:

Coşkun Çelik (2013) Chief Researcher @TUBITAK Uzay (Turkey)
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Dr. Fatih Say (2011) General Manager CEO @EHSİM (Turkey)
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MSc:

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Dilek Baysal (2014)
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Salih Yıldırım (2012) (??)

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Dr. M. İlhan Akbaş (2006) Assoc. Chair & Assoc.Prof. @Embry-Riddle Aeronautical Univ. (USA)
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Dr. Alper Bereketli (2005) Senior Principal AI & Algorithm Engineer @Aselsan (Turkey)
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Dr. Neslihan Bayramoğlu (2005) Docent and Senior Research Fellow @Univ. of Oulu (Finland)
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(<https://www.linkedin.com/in/co%C5%9Fkun-%C3%A7elik-73bb6a99/>)

Dr. Gökhan Özbilgin (2004) Prof. Dr. Mngt. Inf. Sys. (MIS) @Medipol Univ. (Turkey)
(<https://www.linkedin.com/in/izzetgokhanozbilgin/?locale=en>)

Bilal Öğünlü (2004) Senior Engineering Specialist | Freelance Consultant (Dev & Business Planning)
(<https://www.linkedin.com/in/bilal-ogunlu/>)

Kerem Önal (2004) Senior Software Developer @ING (Netherlands)
(<https://www.linkedin.com/in/kerem-onal/>)

Mehmet Nazir Kaçar (2003) Engineer @Matrisled Electronics
(<https://www.linkedin.com/in/mehmet-nazir-ka%C3%A7ar-569945b0/>)

Ahmet Turan Yeler (2002) Engineer @TSK
(<https://www.linkedin.com/in/ahmet-turan-yeler-210893245/>)

Hüseyin Yılmaz (2001) ??
??

Barış Karadeniz (2001) CTO, Founding Partner @Arke Telekom (Turkey)
(<https://www.linkedin.com/in/bar%C4%B1%C5%9F-karadeniz/>)

Erhan Asım Öztürk (2001) Systems Engineering Manager @STM (Turkey)
(<https://www.linkedin.com/in/erhan-as%C4%B1m-%C3%B6zt%C3%BCrk-b9596a44/?locale=en>)

Dilek Ertem (2001) Technical Proposal Manager @Aselsan (Turkey)
(<https://www.linkedin.com/in/dilek-ertem-182457168/>)

Funded Project Work:

Project Manager and Principal Investigator, *Architecture and Performance Benchmarking Studies Phase 2*, METU DSIM Research Project, Project No: 2020-03-01-2-00-001 (with Thales LAS France Inc.), November 2019 – July 2020.

Project Manager and Principal Investigator, *Architecture and Performance Benchmarking Studies*, METU DSIM Research Project, Project No: 2019-03-01-32-00-04 (with Thales LAS France Inc.), November 2018 – August 2019.

Investigator, *Monitoring and recording of low voltage distribution system*, Tübitak 1503 project, Project No: 115E134 (with Aselsan Inc.), February 2016 – March 2019.

Investigator, *Pre-analysis of a defense system (feasibility study for air and missile defense system)* (for Aselsan Inc. via ParGe A.Ş.), January 2015 – September 2015.

Principal Investigator (for comm. part), *Communications infrastructure analysis (within the main project of 'Development of and sensor fusion algorithms*, METU DSIM Project, Project No: SST-IA-2012-97 (for Aselsan Inc.), November 2012 – April 2014.

Principal Investigator, *Pipelined divide-and-conquer multiplier design for finite field $GF(2^m)$ based on irreducible trinomials*, METU Scientific Research Fund, Project No: BAP-03-01-2010-01, February 2010 - January 2011.

Investigator, *Narrowband command and control waveform development - phase 2*, METU DSIM Project, Project No: 09-03-01-2-00-22 (for Aselsan Inc.), 2010-2011.

Investigator, *Narrowband command and control waveform development*, METU DSIM Project, Project No: 08-03-01-2-00-14 (for Aselsan Inc.), 2009-2010.

Investigator, *Domain analysis for weapon systems software and development of a reference software architecture*, METU DSIM Project, Project No: 07-03-02-1-00-19 (for Aselsan Inc.), 2007-2008.

Principal Investigator, *Development of a parallel graph algorithm for the rectilinear Steiner tree problem and its performance evaluation*, METU Scientific Research Fund, Project No: BAP-2008-03-01-06, 1 January 2008 - 31 December 2008.

Administrative Duties:

University Level

Elected Senate Member, İzmir Institute of Technology, Sep. 2023 -

Ethics Committee Chair, İzmir Institute of Technology, Jan. 2025 -

Ethics Committee Chair, İzmir Institute of Technology, Aug. 2023 - Jan. 2025

Ethics Committee Member, İzmir Institute of Technology, Jan. 2022 - Jan. 2025

Research Directorate Executive Board Member, İzmir Institute of Technology, Jan. 2021 - Jan. 2024

Computer and Informatics Committee Member, Middle East Technical University, Mar. 2011 - Feb. 2020.

Faculty Level

Graduate School Board Member, İzmir Institute of Technology, 2021 - 2024.

Faculty Board Member, İzmir Institute of Technology, Engineering Faculty, 2021 - 2024

Faculty Disciplinary Committee Member, Middle East Technical University, Engineering Faculty, Sep 2018 - Feb 2020.

Faculty Executive Board Member, Middle East Technical University, Engineering Faculty, May 2005 - May 2008.

Department Level

Minor Program Committee Member, İzmir Institute of Technology, Engineering Faculty, Computer Eng. Dep., Jan. 2025 -

Chair, İzmir Institute of Technology, Engineering Faculty, Computer Eng. Dep., 2021 - 2024.

Education Committee Member, İzmir Institute of Technology, Engineering Faculty, Computer Eng. Dep., Jan. 2021 - Nov. 2021

Ethics Committee Member, Middle East Technical University, Engineering Faculty, EE Eng. Dep., Oct. 2019 - Feb. 2020.

Undergraduate Program Committee Member, Middle East Technical University, Engineering Faculty, EE Eng. Dep., Oct. 2017 – Feb. 2020.

Coordinator for Information Technology Affairs, Middle East Technical University, Engineering Faculty, EE Eng. Dep., Nov. 2014 – Sep. 2019.

Academic Recruitment Committee Member, Middle East Technical University, Engineering Faculty, EE Eng. Dep., Sep. 2013 – Sep. 2016.

Undergraduate Program Committee Member, Middle East Technical University, Engineering Faculty, EE Eng. Dep., June 2011 – Sep. 2013.

Student Performance Evaluation Committee Member, Middle East Technical University, Engineering Faculty, EE Eng. Dep., Mar. 2009 – Mar. 2015.

Vice Chair, Middle East Technical University, Engineering Faculty, EE Eng. Dep., Oct. 2005-Dec. 2007.

Budget and Planning Committee Member, Middle East Technical University, Engineering Faculty, EE Eng. Dep., 2003-2004.

Undergraduate Program Committee Member, Middle East Technical University, Engineering Faculty, EE Eng. Dep., 2001-2002.

Publicity and Public Relations Committee Member, Middle East Technical University, Engineering Faculty, EE Eng. Dep., 1999-2000.

Computer Center Departmental Coordinator, Middle East Technical University, Engineering Faculty, EE Eng. Dep., 2000-2005.

Scientific, Vocational and Social Memberships:

Senior Member, Institute of Electrical and Electronics Engineering (IEEE)

Member, IEEE Computer Society

Member, IEEE Communications Society

Member, Association for Computing Machinery (ACM)

Member, Operational Research Society (ORS)

Member, Chamber of Electrical Engineers (EMO)

Member, METU Alumni Society (OMD)

Member, UMIST Alumni Society

Honors, Awards and Scholarships:

“2014 Annual High Performance Award”, Middle East Technical University.

Supervisor of the “Best Thesis of the Year Award 2010-2011”, Middle East Technical University, Graduate School of Natural and Applied Sciences.

PhD Scholarship, Higher Education Council, Turkey, 1992-1996.

University Admissions Exam, High Achievement Award

Undergraduate Education Scholarship - High Achievement Award in the University Admissions Exam
- Sabancı Holdings PLC, Turkey, 1984-1988.

Taught Courses (Undergraduate and graduate level):

Undergraduate

CENG322 Operating Systems (Comp. Eng. Dep., IZTECH)
Semester: 2025-26/2

CENG311 Computer Architecture (Comp. Eng. Dep., IZTECH)
Semester: 2023-24/1, 2024-25/1, 2025-26/1

CENG312 Computer Networks (Comp. Eng. Dep., IZTECH)
Semester: 2022-23/2, 2023-24/2, 2024-25/2, 2025-26/2

CENG115 Discrete Structures (Comp. Eng. Dep., IZTECH)
Semester: 2023-24/1

CENG112 Data Structures (Comp. Eng. Dep., IZTECH)
Semester: 2019-20/2, 2020-21/2, 2021-22/2, 2022-23/2

CENG451 Advanced Digital Systems (Comp. Eng. Dep., IZTECH)
Semester: 2020-21/1, 2021-22/1

EE445 Computer Architecture I (EE Eng. Dep., METU)
Semester: 2017-18/1, 2018-19/1, 2019-20/1

EE446 Computer Architecture II (EE Eng. Dep., METU)
Semester: 2017-18/2, 2018-19/2

EE442 Operating Systems (EE Eng. Dep., METU)
Semester: 2016-17/2, 2017-18/2, 2018-19/2

EE348 Logic Design (EE Eng. Dep., METU)
Semester: 1996-97/2, 1996-97/3 (summer), 1998-99/2, 1999-00/2,
2000-01/2, 2001-02/2, 2002-03/2, 2003-04/2, 2003-04/3
(summer), 2004-05/2, 2005-06/2, 2006-07/2, 2007-08/2,
2009-10/2, 2010-11/2, 2011-12/2, 2012-13/2, 2013-14/2,
2014-15/2, 2015-16/2

EE441 Data Structures (EE Eng. Dep., METU)
Semester: 1999-00/1, 2000-01/1, 2001-02/1, 2005-06/1, 2014-15/1

EE447 Microprocessors (EE Eng. Dep., METU)
Semester: 1997-98/1, 1999-00/1, 2000-01/1, 2001-02/1, 2002-03/1,
2003-04/1, 2004-05/1, 2006-07/1, 2007-08/1, 2008-09/1,
2009-10/1, 2010-11/1, 2011-12/1, 2012-13/1, 2013-14/1

EEE347/CENG336 Microprocessors/ Embedded System Design (Electrical
and Electronics Eng. Dep., METU Northern Cyprus Campus)
Semester: 2006-07/2, 2008-09/2, 2011-12/2, 2012-13/2

EEE248/CNG232 Logic Design (EE Eng. Dep., METU Northern Cyprus Campus)
Semester: 2008-09/2

EE314 Digital Electronics Laboratory (EE Eng. Dep., METU)
Semester: 2003-04/2, 2004-05/2, 2005-06/2

EE444 Computer Networks (EE Eng. Dep., METU)
Semester: 1999-00/2, 2000-01/2, 2001-02/2, 2002-03/2, 2003-04/2,
2004-05/2, 2005-06/2

BIL256 Data Communications (Computer Eng. Dep., Başkent University)
Semester: 2000-01/2

EE312 Digital Electronics (EE Eng. Dep., METU)
Semester: 1996-97/2, 1998-99/2

Graduate

CENG504 Optimization Methods (Comp. Eng. Dep., IZTECH)
Semester: 2020-21/2, 2022-23/1, 2023-24/2, 2024-25/2, 2025-26/2

CENG505 Advanced Computer Networks (Comp. Eng. Dep., IZTECH)
Semester: 2020-21/1, 2021-22-1, 2022-23/1, 2023-24/1, 2024-25/1

CENG515 Special Topics in Computer Science (Comp. Eng. Dep., IZTECH)
Semester: 2019-20/2

EE546 Automata Theory II (EE Eng. Dep., METU)
Semester: 2007-8/2 2009-10/2, 2010-11/2, 2011-12/2, 2012-13/2,
2013-14/2, 2014-15/1, 2015-16/1, 2016-17/1, 2017-18/1,
2018-19/1, 2019-20/1

EE545 Automata Theory I (EE Eng. Dep., METU)
Semester: 2007-8/1, 2008-9/1, 2009-10/1, 2010-11/1, 2011-12/1,
2012-13/1, 2013-14/1

EE544 Algorithms and Comp. Complexity (EE Eng. Dep., METU)
Semester: 2004-5/1, 2005-6/1

EE703 Advanced Data Structures and Algorithms (EE Eng. Dep., METU)
Semester: 1997-8/1, 2001-2/2

Laboratory Development and Supervision:

Microcomputer Systems Laboratory: The Department of Electrical and Electronics Engineering, METU, used MC6800-based microcomputer devices and training kits for microprocessor and computer architecture related courses before 2000. Creating appropriate funding, these labs have been modernized equipped with new laptop computers, Motorola (later Freescale) 6811 based training sets and other necessary support equipment in 2001. A completely new set of lecture notes and laboratory manuals were prepared and labs were managed and supervised by Dr. Bazlamaçcı afterwards.

Starting in 2016, the associated courses and laboratories are modernized again, re-organized and enhanced using ARM and FPGA based training sets and the necessary support equipment.

Computer Systems and Networks Research Laboratory: This laboratory was established in Electrical and Electronics Engineering Department in 2002 as a graduate research laboratory. It was equipped with personal computers, work stations, a variety of advanced microcomputer development environments, DSP cards, network simulation software, etc. Many undergraduate and graduate students have participated in research activities of the laboratory under the supervision of Dr. Bazlamaçcı.

Conference and Symposium Organization:

Advisory Board Member / Program Committee Member, 10th Int. Conf. on Computer Science and Eng. (UBMK 2026), İstanbul, Turkey

Program Committee Member, Signal Processing and Communications Applications Conf. (SIU 2026), İstanbul, Turkey.

Program Committee Member, 10th Int. Conf. on Computer Science and Eng. (UBMK 2025), İstanbul, Turkey

Advisory Board Member, 9th Int. Conf. on Computer Science and Eng. (UBMK 2024), Antalya, Turkey

Advisory Board Member, 8th Int. Conf. on Computer Science and Eng. (UBMK 2024), Burdur, Turkey

Advisory Board Member, 7th Int. Conf. on Computer Science and Eng. (UBMK 2024), Diyarbakır, Turkey

Program Committee Member, Processor Design Workshop, 19 September 2019, İstanbul, Turkey

Program Committee Member, Embedded Systems and Applications Symposium (GÖMSİS'14), 4-5 December 2014, İstanbul, Turkey

Program Committee Member, Embedded Systems and Applications Symposium (GÖMSİS'12), 29-30 November 2012, İstanbul, Turkey

Program Committee Member, 4th Nat. Symp. on Electrical Electronics Computer Biomedical Eng. Education, 22-24 October 2009, Eskişehir, Turkey

Proceedings Editor, 1st Nat. Symp. on Electrical Electronics Computer Eng. Education, 30 April-2 May 2003, Ankara, Turkey.

Advisory, Program and Organizing Committee Member, 1st Nat. Symp. on Electrical Electronics Computer Eng. Education, 30 April-2 May 2003, Ankara, Turkey.

Advisory Committee Member, 1st Nat. Symp. on Communication Technologies 17-21 October 2001, Ankara, Turkey.

Refereeing and Jury Memberships:

IZTECH MSc and PhD Thesis Evaluation Committees, **Jury Member**

METU MSc and PhD Thesis Evaluation Committees, **Jury Member**

IV. TÜBİTAK – TTGV – TUSİAD Technology Awards Program, **Jury Member**

TÜBİTAK - BAYG High School Research Projects Competition, **Jury Member**

Marine Forces New Innovations Project Competition, **Jury Member**

METU Technopolis New Ideas New Business Project Competition, **Jury Member**

Mustafa Parlar Vakfı Science Award, **Jury Member**

TÜBİTAK- TİDEB –TEYDEB Projects, **Referee**

TÜBİTAK- Bilten Projects, **Referee**

METU Technopolis Projects, **Referee**

METU Kosgeb Projeleri, **Referee**

METU Engineering Faculty, School of Science Institute, Informatics Institute, Institute of Applied Mathematics, AFP-BAP Projects, **Referee**

IEEE Transactions on Computers, **Reviewer**

IEEE Transactions on Very Large Scale Integrated Systems, **Reviewer**

IEEE Transactions on Circuits and Systems, **Reviewer**

Computer Networks Journal, Elsevier, **Reviewer**

Computers & Operations Research, Elsevier, **Reviewer**

Journal of Heuristics, **Reviewer**

Elektrik - Turkish J. of Electr. Eng. & Comp. Sciences, **Reviewer**

Elektrik - Turkish J. of Electr. Eng. & Comp. Sciences, **Editorial Board Member**

ARI The Bulletin of the Istanbul Tech. Uni. - An Interdisciplinary J. of Physical and Eng Science, **Reviewer**

Doğuş University Journal, **Reviewer**

Anadolu University Science and Technology Journal, **Reviewer**

Gazi University Engineering and Architecture Faculty Journal, **Reviewer**

Niğde University Engineering Sciences Journal, **Reviewer**

ISCIS Int. Symp. on Comp. and Information Sciences, **Reviewer**

ISCTurkey Int. Information Security and Cryptology Conference, **Reviewer**

SAVTEK Defence Technologies Congress, **Reviewer**

ITUS Communications Technologies National Symp., **Reviewer**

ASES Avionics and Systems Integration Symp., **Reviewer**
PDP Euromicro Int. Conf. Parallel, Distr. and N/w-based Processing, **Reviewer**
GÖMSİS Embedded Systems and Applications Symp., **Reviewer**
SİU Signal Processing and Communications Applications Conf., **Reviewer**

TUBITAK-TIDEB-TEYDEB Projects, **Supervisor**
METU Technopolis Projects, **Supervisor**
METU Kosgeb Projects, **Supervisor**
İzmir Technopark Projects, **Reviewer**

TUBITAK Projects, **Review Panel member**
Turkish Republic Ministry of Industry, San-Tez Projects, **Review Panel member**
TUBITAK Embedded Systems, **Call Program Board Member**

Turkish Republic Justice Department, **Expert Reporter**

WoS Record:

Document Count: 21 Citation Count: 117 H-index: 6